

Part A: (35 points)

1. (15 points)

```
setwd("/Users/shrwydbrg/Desktop")  
Signals=read.csv(file="/Users/shrwydbrg/Desktop/R/DataSets/Signals_A_62016.csv",header=TRUE,as.is=TRUE,na.strings=c("NA",".", ""))
```

```
Signals$PFormDate=as.Date(Signals$PFormDate,format = "%d/%m/%Y")  
Signals$CompInd=ifelse(Signals$BBHL=="Computers",1,0)  
describe(Signals$PermNo)  
describe(Signals$PFormDate)  
describe(Signals$BBHL)  
describe(Signals$CompInd)
```

119463 observations, 8859 unique firms, 53 trading portfolio formation dates, and 18 BBHL industries.

```
describe((Signals$PermNo)[Signals$CompInd=="1"])
```

15770 observations, and 1582 unique firms.

2. (20 points)

```
CrossSecDF=Signals[Signals$PFormDate=="1992-06-30",]  
CrossSecDF=CrossSecDF[,c("PFormDate","Size","BTM","AvgMonRet","ETP","PermNo","CompInd","PFormDate")]  
CrossSecDF_New=CrossSecDF[!is.na(CrossSecDF$ETP)&!is.na(CrossSecDF$BTM)&!is.na(CrossSecDF$Size)&!is.na(CrossSecDF$AvgMonRet)],]  
describe((CrossSecDF_New$PermNo)[CrossSecDF_New$CompInd=="1"])
```

238 observations, and 238 unique firms.

```
describe((CrossSecDF_New$PermNo)[CrossSecDF_New$CompInd=="0"])
```

1688 observations, and 1687 unique firms.

Part B: (65 points)

3. (30 points)

```
CrossSecDF_New$logSize=log(CrossSecDF_New$Size)  
CrossSecDF_New$logBTM=log(CrossSecDF_New$BTM)
```

Equation 1: $AvgMonRet = \beta_0 + \beta_1 \log(Size) + \beta_2 \log(BTM) + \beta_3 ETP + U$

Fit= lm(AvgMonRet~logSize+logBTM+ETP, data = CrossSecDF_New/CrossSecDF_Computers/ CrossSecDF_Others

```
CrossSecDF_Computers=CrossSecDF_New[Signals$CompInd=="1",]  
CrossSecDF_Others=CrossSecDF_New[Signals$CompInd=="0",]
```

The results are summarized in Table 1.

when we test for homoscedasticity, we can reject the null hypothesis for all the models. We will conclude that the errors are not homoscedastic and we will use the HCO variance-covariance matrix in order to calculate the confidence intervals.

```
bptest(FIT_combined,studentize = FALSE)
bptest(FIT_Computers,studentize = FALSE)
bptest(FIT_Others,studentize = FALSE)
vcovHCO=vcovHC(FIT_combined,type="HCO")
coefci (FIT_combined,vcov.=vcovHCO,
"logBTM",level=0.95,alternative="two.sided")
vcovHCO=vcovHC(FIT_Computers,type="HCO")
coefci (FIT_Computers,vcov.=vcovHCO,
"logBTM",level=0.95,alternative="two.sided")
vcovHCO=vcovHC(FIT_Others,type="HCO")
coefci (FIT_Others,vcov.=vcovHCO, "logBTM",level=0.95,alternative="two.sided")
```

The results are summarized in Table 2.

The downside of using separate estimates is that it does not allow us to conduct direct comparisons of the parameters of interest or joint hypotheses tests about population parameters. Instead, we get separate estimates from separate samples.

In order to find interval estimate for the difference between skewness of errors for firms that belong to Computers industry and firms that belong to other industries, we will use the bootstrap approach.

```
residuals = cbind(FIT_Others$residuals, FIT_Computers$residuals)
orig_sample = residuals
n = dim(residuals)[1]
orig_index = 1:n
generate the Bootstrap sampling distribution of the estimator
boot_estimates = rep(NA, times=10000)
for(i in 1:10000){
  boot_index = sample(orig_index, n, replace=TRUE)
  boot_sample = orig_sample[boot_index, ]
  boot_estimates[i] = skewness((boot_sample)[ ,1])- skewness((boot_sample)[ ,2])
}
se_boot = sd(boot_estimates)
orig_sample_estimate = skewness((boot_sample)[ ,1])- skewness((boot_sample)[ ,2])
```

Using the standard error method:

$$-5.378965 < \widehat{\text{skewness}}_{\text{Others}} - \widehat{\text{skewness}}_{\text{CompIn}}$$

```
CI_95_se_method = c(orig_sample_estimate + qnorm(p=0.025, mean=0,
sd=1)*se_boot,
```

```
orig_sample_estimate - qnorm(p=0.025, mean=0, sd=1)*se_boot)
```

Using the percentile method

$$-0.7730944 < \widehat{\text{skewnessOthers}} - \widehat{\text{skewnessCompInd}} < 6.7925200$$

```
qs_boot = quantile(boot_estimates, probs=c(0.025, 0.975))
```

```
CI_95_p_method = unname(qs_boot)
```

4. (19 points)

$$\text{Equation 2: } \text{AvgMonRet} = \beta_0 + \beta_1 \log(\text{Size}) + \beta_2 \log(\text{BTM}) + \beta_3 \text{ETP} + \beta_4 \text{CompInd} + \beta_5 \text{CompInd} \times \log(\text{BTM}) + U$$

FIT2 = lm(AvgMonRet~logSize+logBTM+ETP+CompInd+CompInd*logBTM, data = CrossSecDF_New)

The results are summarized in Table 1.

Table 1: Regression results for Equations 1(combined & by industry) and Equation 2.

	Dependent variable:			
	AvgMonRet			
	(Combined)	(Computer)	(Others)	(FIT2)
logSize	-0.230*** (0.048)	-0.065 (0.111)	-0.256*** (0.052)	-0.212*** (0.048)
logBTM	0.475*** (0.107)	0.599** (0.284)	0.472*** (0.116)	0.590*** (0.115)
ETP	-0.551 (0.383)	-3.754 (4.659)	-0.549 (0.393)	-0.581 (0.383)
CompInd				0.292 (0.378)
logBTM: CompInd				-0.572* (0.304)
Constant	3.613*** (0.257)	2.790*** (0.673)	3.793*** (0.284)	3.503*** (0.263)
Observations	1,922	287	1,635	1,922
R ²	0.029	0.020	0.031	0.035
Adjusted R ²	0.027	0.009	0.030	0.032
Residual Std. Error	3.887 (df = 1918)	3.360 (df = 283)	3.972 (df = 1631)	3.877 (df = 1916)
F Statistic	18.956*** (df = 3; 1918)	1.909 (df = 3; 283)	17.682*** (df = 3; 1631)	13.819*** (df = 5; 1916)

According to this table, point estimators are as follow:

$$\hat{\beta}_0 = 3.50305, \hat{\beta}_1 = -0.21244, \hat{\beta}_2 = 0.59032, \hat{\beta}_3 = -0.58129, \hat{\beta}_4 = 0.29194, \hat{\beta}_5 = -0.57229.$$

- For a firm which is not part of the computer industry, a 1% increase in the BTM of a firm decreases the Average monthly return of a firm's stock by 0.0059 ($\beta_2/100$)

- For a firm which is part of the computer industry, a 1% increase in the BTM of a firm decreases the Average monthly return of a firm's stock by 0.0059-0.0057 ($(\beta_2 + \beta_5)/100$)
- For companies which are part of the computer industry there is increase of 0.29194 in the Average monthly return.

```
bptest(FIT2_combined,studentize = FALSE)
```

H0: $\beta_2=0$

H1: otherwise

```
vcovHCO=vcovHC(FIT2_combined,type="HCO")
```

```
linearHypothesis(FIT2_combined,"logBTM=0",vcov.=vcovHCO)
```

For the first hypothesis, the p-value equals to 9.431e-07, we will reject the null hypothesis and conclude that β_2 is significantly different from zero.

H0: $\beta_2 + \beta_5=0$

H1: otherwise

```
linearHypothesis(FIT2_combined,c("logBTM+logBTM:CompInd=0"),test="Chisq",v
```

```
cov.=vcovHCO)
```

For the SECOND hypothesis, the p-value equals to 0.538, we will NOT be able to reject the null hypothesis, thus, we will conclude that $\beta_2 + \beta_5$ is not significantly different from zero.

5. (16 points)

Equation 3: $CompInd = \beta_0 + \beta_1 \log(Size) + \beta_2 \log(BTM) + \beta_3 ETP + \beta_4 AvgMonRet + U$

```
FIT_OLS=lm(CompInd~logSize+logBTM+ETP+AvgMonRet, data =CrossSecDF_New)
```

Equation 4: $P(CompInd = 1|X) = G(\beta_0 + \beta_1 \log(Size) + \beta_2 \log(BTM) + \beta_3 ETP + \beta_4 AvgMonRet)$

```
FIT_Logit=glm(CompInd~logSize+logBTM+ETP+AvgMonRet, data =CrossSecDF_New)
```

$G(\cdot)$ is the c.d.f. of a standard logistic random variable

The results are summarized in Table 3.

Table 3: Regression results for Equations 3 and 4

	Dependent variable:	
	Comp_Ind	
	OLS	logistic
logSize	-0.019*** (0.004)	-0.178*** (0.040)
logBTM	-0.052*** (0.009)	-0.406*** (0.085)
ETP	0.002 (0.032)	-1.092 (1.661)
AvgMonRet	0.006*** (0.002)	0.039** (0.015)
Constant	0.174*** (0.023)	-1.401*** (0.254)

Observations	1,922	1,922
R ²	0.028	
Adjusted R ²	0.025	
Log Likelihood		-694.876
Akaike Inf. Crit.		1,399.752
Residual Std. Error	0.325 (df = 1917)	
F Statistic	13.563*** (df = 4; 1917)	
